

LENGHU, China

WMO: 526020

Lat: 38.8283N

Lon: 93.3787E

Elev: 2771

StdP: 72.17

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -22.3	(c) -20.6	(d) -35.5	(e) 0.2	(f) -12.3	(g) -33.1	(h) 0.2	(i) -12.4	(j) 14.3	(k) -8.2	(l) 12.6	(m) -8.9	(n) 1.0	(o) 320	(p) 0.703

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 12.8	(c) 28.1	(d) 11.0	(e) 26.4	(f) 10.0	(g) 24.9	(h) 9.3	(i) 12.7	(j) 24.1	(k) 11.6	(l) 22.6	(m) 10.5	(n) 22.0	(o) 4.5	(p) 290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 8.0	(b) 9.4	(c) 17.5	(d) 6.4	(e) 8.4	(f) 17.1	(g) 4.7	(h) 7.4	(i) 16.7	(j) 45.4	(k) 24.2	(l) 41.5	(m) 22.6	(n) 38.5	(o) 22.1	(p) 16.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.1	11.3	9.7	-26.0	31.0	2.0	1.9	-27.4	32.4	-28.6	33.5	-29.8	34.6	-31.3	36.0
			-26.6	13.7	1.9	1.2	-27.9	14.5	-29.0	15.2	-30.1	15.9	-31.5	16.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	4.1	-10.8	-6.7	-1.5	4.9	10.4	15.2	18.2	17.2	11.5	3.5	-4.1	-9.8	(6)
(7)		DBStd	10.63	3.23	3.87	3.91	3.98	3.12	2.56	2.84	2.73	3.04	3.42	3.60	2.89	(7)
(8)		HDD10.0	2919	645	468	357	157	33	1	0	0	16	203	424	614	(8)
(9)		HDD18.3	5268	903	702	615	402	246	97	37	55	205	461	674	873	(9)
(10)		CDD10.0	749	0	0	0	6	46	158	254	223	62	1	0	0	(10)
(11)		CDD18.3	57	0	0	0	0	0	4	33	19	1	0	0	0	(11)
(12)		CDH23.3	717	0	0	0	1	6	68	383	247	13	0	0	0	(12)
(13)		CDH26.7	132	0	0	0	0	0	4	85	42	1	0	0	0	(13)
(14)	Wind	WSAvg	3.6	2.4	3.4	4.1	4.3	4.3	4.2	4.0	3.6	3.4	2.6	2.2	(14)	
(15)	Precipitation	PrecAvg	19	1	0	1	1	1	4	3	3	0	0	0	(15)	
(16)		PrecMax	38	7	3	2	7	11	28	15	13	20	3	0	2	(16)
(17)		PrecMin	7	0	0	0	0	0	0	0	0	0	0	0	0	(17)
(18)		PrecStd	8	1	1	1	1	3	6	3	4	5	1	0	1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.6	7.9	14.9	20.9	24.1	27.1	31.6	30.1	25.1	18.1	9.6	3.2	(19)
(20)			MCWB	-4.4	-1.6	2.2	5.5	7.0	9.6	13.4	12.0	9.1	4.5	0.1	-3.6	(20)
(21)		2%	DB	-0.2	5.4	11.6	18.1	21.8	25.1	28.9	27.7	22.6	15.6	7.7	0.4	(21)
(22)			MCWB	-6.1	-2.9	0.5	4.1	6.2	8.8	11.4	10.6	7.9	3.3	-1.2	-5.5	(22)
(23)		5%	DB	-1.9	3.1	9.3	16.1	20.2	23.8	27.1	26.0	20.8	13.5	5.9	-1.3	(23)
(24)			MCWB	-7.2	-4.1	-0.7	3.3	5.5	8.1	10.6	9.8	6.8	2.3	-2.2	-6.6	(24)
(25)		10%	DB	-3.8	1.1	7.2	14.1	18.6	22.5	25.4	24.4	19.0	11.5	3.8	-3.0	(25)
(26)			MCWB	-8.4	-5.3	-1.7	2.2	4.9	7.7	9.9	9.3	5.9	1.3	-3.3	-7.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-4.4	-1.3	2.4	5.8	7.7	11.7	14.5	13.8	11.0	5.0	0.3	-3.6	(27)
(28)			MCDWB	2.2	7.2	14.4	20.0	21.8	21.8	28.7	25.4	20.7	16.4	9.1	2.6	(28)
(29)		2%	WB	-6.0	-2.7	0.7	4.5	6.8	10.2	13.1	12.6	9.0	3.6	-1.1	-5.4	(29)
(30)			MCDWB	-0.4	5.1	11.0	17.5	19.9	20.7	25.1	23.2	19.5	14.7	7.3	0.1	(30)
(31)		5%	WB	-7.1	-4.0	-0.5	3.4	6.0	9.3	12.0	11.4	7.8	2.6	-2.1	-6.5	(31)
(32)			MCDWB	-2.1	3.0	8.9	15.5	18.8	20.7	23.7	21.7	18.5	13.0	5.8	-1.5	(32)
(33)		10%	WB	-8.4	-5.3	-1.6	2.4	5.2	8.5	10.9	10.4	6.7	1.5	-3.2	-7.7	(33)
(34)			MCDWB	-3.9	0.9	7.0	13.6	17.5	20.2	22.9	21.8	17.2	11.1	3.6	-3.1	(34)
(35)	Mean Daily Temperature Range	MDBR		15.2	15.5	15.8	15.6	14.3	13.1	12.8	13.5	14.4	15.6	15.6	14.9	(35)
(36)		5% DB	MCDBR	16.4	16.8	18.1	17.3	16.4	15.1	14.7	15.3	15.6	16.9	16.7	15.9	(36)
(37)			MCWBR	11.8	11.0	10.4	8.6	7.6	5.9	5.2	5.5	6.7	8.5	10.0	11.3	(37)
(38)		5% WB	MCDBR	16.0	16.4	17.4	16.3	14.5	11.9	11.9	11.7	12.8	16.0	16.3	15.4	(38)
(39)			MCWBR	11.5	10.7	10.0	8.2	6.8	4.9	4.4	4.6	5.6	8.2	9.8	11.0	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.249	0.293	0.352	0.416	0.431	0.442	0.429	0.403	0.368	0.310	0.270	0.245	(40)
(41)		taud		2.345	2.191	2.001	1.840	1.861	1.902	2.001	2.059	2.114	2.212	2.298	2.365	(41)
(42)		Ebn at Noon		941	927	896	857	851	839	847	861	870	893	899	920	(42)
(43)		Edh at Noon		87	118	160	201	201	194	174	160	142	114	91	79	(43)
(44)	All-Sky Solar Radiation	RadAvg		2.93	3.96	5.43	6.61	7.32	7.24	6.98	6.63	5.67	4.62	3.32	2.66	(44)
(45)		RadStd		0.18	0.16	0.17	0.25	0.27	0.40	0.30	0.27	0.25	0.14	0.12	0.11	(45)

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-133	+61	N/A
(47)	Regional (1 neighbor)	+0.33	N/A	N/A	N/A	N/A	N/A	N/A	-106	+84	+31

Nomenclature: See separate page